

Task 2 Final Report

Deep Learning Text Classification (Sentiment Analysis)

1. Objective

Develop a deep learning model to classify movie reviews as positive or negative. The project demonstrates the complete NLP workflow including preprocessing, tokenization, model training, evaluation, visualization, and model saving.

2. Problem Statement

Sentiment analysis helps organizations understand customer opinions from textual data. This project predicts review sentiment using a Long Short-Term Memory (LSTM) neural network.

3. Dataset

The IMDB Movie Reviews dataset from TensorFlow/Keras was used. It contains 50,000 labeled reviews with balanced positive and negative sentiment classes.

4. Data Preprocessing

- Loaded the IMDB dataset.
- Used tokenization through the IMDB word index.
- Converted reviews into integer sequences.
- Applied sequence padding to a fixed length.
- Created training, validation, and testing datasets.

5. Model Architecture

The deep learning model consists of an Embedding layer, an LSTM layer, a Dropout layer, a Dense hidden layer with ReLU activation, and a Sigmoid output layer for binary classification.

6. Training

The model was trained using the Adam optimizer and Binary Cross-Entropy loss. Early Stopping was used to reduce overfitting and restore the best model weights.

7. Evaluation

Performance was evaluated using Accuracy, Precision, Recall, F1-Score, Confusion Matrix, Training & Validation Accuracy/Loss Curves, and inference on unseen reviews.

8. Results

The LSTM model achieved strong sentiment prediction performance. Training and validation curves showed effective learning, and evaluation metrics demonstrated reliable classification.

9. Tools

Python, Google Colab, TensorFlow, Keras, NumPy, Pandas, Matplotlib, Scikit-learn.

10. Learning Outcomes

- Learned NLP preprocessing.
- Applied tokenization and padding.
- Built an LSTM model.
- Evaluated a deep learning classifier.
- Saved the trained model for future inference.

11. Future Scope

Future enhancements include using pretrained embeddings (GloVe), transformer models such as BERT/DistilBERT, hyperparameter tuning, and deployment using Flask, FastAPI, or Streamlit.

12. Conclusion

This project successfully fulfilled the internship requirements for a deep learning text classification task by implementing an end-to-end sentiment analysis solution.

13. Internship Requirements Checklist

Requirement	Status
Deep Learning Model	Completed
Text Classification	Completed
Tokenization	Completed
LSTM Model	Completed
Training Curves	Completed
Evaluation Metrics	Completed
Confusion Matrix	Completed
Saved Model	Completed
Inference	Completed